L3-Searching and Sorting

Algorithm	Best	Reason	Average	Reason	Worst	Reason	Space	Reason
Linear Search (Normal)	O(1)	First element-le kidaikkum.	O(n)	Half array check pannuvom.	O(n)	Ella elements check pannuvom.	O(1)	Extra variable mattum.
Linear Search (Recursion)	O(1)	First call-le kidaikkum.	O(n)	Half recursive calls.	O(n)	Ella recursive calls pogum.	O(n)	Recursion stack use aagum.
Binary Search (Normal)	O(1)	Middle-le kidaikkum.	O(log n)	Half-half search pannuvom.	O(log n)	Half-half reduce pannuvom.	O(1)	Extra variable mattum.
Binary Search (Recursion)	O(1)	First middle-le kidaikkum.	O(log n)	Half-half recursion.	O(log n)	Last recursion vara pogum.	O(log n)	Recursion stack.
Selection Sort	$O(n^2)$	Ella pass-um compare pannuvom.	$O(n^2)$	Ella pass-um compare pannuvom.	$O(n^2)$	Ella pass-um compare pannuvom.	O(1)	Extra array illa.
Bubble Sort	O(n)	Already sorted.	$O(n^2)$	Neraya swaps.	$O(n^2)$	Maximum swaps.	O(1)	Extra array illa.
Insertion Sort	O(n)	Already sorted.	$O(n^2)$	Elements shift aagum.	$O(n^2)$	Maximum shifting.	O(1)	One temp variable.
Shell Sort	O(n log n)	Gap use pannuvom.	O(n^1.5)	Gap depend pannum.	$O(n^2)$	Bad gap sequence.	O(1)	Extra array illa.
Merge Sort	O(n log n)	Divide & Merge.	O(n log n)	Divide & Merge.	O(n log n)	Divide & Merge.	O(n)	Temporary array use pannuvom.
Quick Sort	O(n log n)	Good pivot.	O(n log n)	Balanced partition.	$O(n^2)$	Bad pivot.	O(log n) Avg / O(n) Worst	Recursion stack.
Counting Sort	O(n+k)	Counting pannuvom.	O(n+k)	Counting pannuvom.	O(n+k)	Counting pannuvom.	O(n+k)	Count array use pannuvom.
Radix Sort	O(d(n+k))	Digit by digit sort.	O(d(n+k))	Digit by digit sort.	O(d(n+k))	Digit by digit sort.	O(n+k)	Count array use pannuvom.
Heap Sort	O(n log n)	Heap use pannuvom.	O(n log n)	Heap use pannuvom.	O(n log n)	Heap use pannuvom.	O(1)	Extra array illa.
Bucket Sort	O(n+k)	Equal buckets.	O(n+k)	Equal buckets.	$O(n^2)$	Oru bucket-la ellam pogum.	O(n+k)	Bucket array use pannuvom.

Stable-na Enna?
Definition:
Same value irukkura elements oda **original order change aagadhu**.
Example:
Before:
5A 3 5B 2

After Stable Sort:
2 3 5A 5B ☑
5A first vandhadhu, **5B** apram vandhadhu. Order same.

In-place-na Enna?
Definition:
Extra array use pannama, original array-la dhaan sorting pannuvom.
Example:
Original Array
↓

Swap pannuvom

↓

Same Array-la sort aagum
Extra array illa.

Algorithm	Stable	In-place
Linear Search	Not Applicable	Yes
Binary Search	Not Applicable	Yes
Selection Sort	❌ No	✅ Yes
Bubble Sort	✅ Yes	✅ Yes
Insertion Sort	✅ Yes	✅ Yes
Shell Sort	❌ No	✅ Yes
Merge Sort	✅ Yes	❌ No
Quick Sort	❌ No	✅ Yes
Counting Sort	✅ Yes	❌ No
Radix Sort	✅ Yes	❌ No
Heap Sort	❌ No	✅ Yes
Bucket Sort	✅ Yes *(usually)*	❌ No

Algorithm	Stable	Reason	In-place	Reason
Selection Sort	❌ No	Swap nala equal elements order change aagum.	✅ Yes	Extra array use panna maatom.
Bubble Sort	✅ Yes	Equal elements swap panna maatom.	✅ Yes	Original array-la sort pannuvom.
Insertion Sort	✅ Yes	Equal elements order maintain aagum.	✅ Yes	One temp variable mattum use pannuvom.
Shell Sort	❌ No	Gap swap nala order change aagum.	✅ Yes	Original array-la sort pannuvom.
Merge Sort	✅ Yes	Merge pannumbodhu order maintain pannuvom.	❌ No	Temporary array use pannuvom.
Quick Sort	❌ No	Pivot swap nala order change aagum.	✅ Yes	Original array-la partition pannuvom.
Counting Sort	✅ Yes	Equal elements order maintain pannum.	❌ No	Count array + Output array use pannuvom.
Radix Sort	✅ Yes	Stable Counting Sort use pannum.	❌ No	Extra Count & Output array use pannuvom.
Heap Sort	❌ No	Heapify swap nala order change aagum.	✅ Yes	Original array-la heap build pannuvom.
Bucket Sort	✅ Yes *(Usually)*	Bucket-kulla order maintain pannalaam.	❌ No	Extra buckets use pannuvom.